

R&D DOCUMENT ON SITE-TO-SITE SETUP USING HYPER-V

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PURPOSE :

Establish a secure Site-to-Site VPN tunnel between an on-premises Hyper-V host (running Windows and RRAS) and an Azure Virtual Network via Azure VPN Gateway (IPsec/IKE).

REQUIRED ENVIRONMENT

Hyper-V (On-Premises):

- Host machine running Windows Server or Windows 10/11 Pro with Hyper-V role enabled.
- A virtual machine on Hyper-V representing the VPN/RRAS server.
- Two virtual switches:
 - Internal switch, for LAN-facing (e.g. your virtual LAN).
 - External switch, bound to host's physical NIC for Internet access.
- RRAS (Routing and Remote Access Service) role installed and configured.

Azure:

- Active Azure subscription.
- A Virtual Network (VNet) with suitable CIDR (non-overlapping with on-prem LAN).
- Gateway Subnet defined (/27 or larger).

- VPN Gateway (route-based) deployed.
- A Local Network Gateway object representing your on-prem VPN endpoint.
- A VPN connection set up with shared key.

STEP-BY-STEP CONFIGURATION

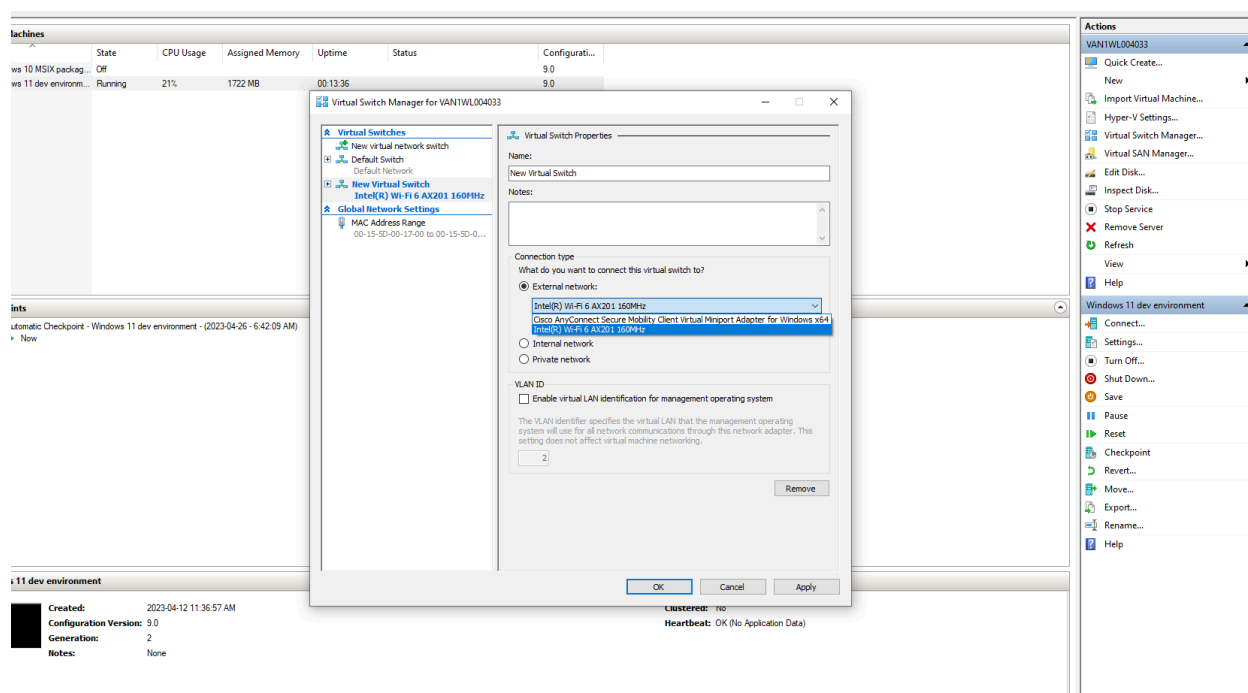
A. Hyper-V & RRAS Setup

1. Enable the Hyper-V feature and create two virtual switches:
 - External switch for WAN access.
 - Internal switch for your internal network where test VMs reside.
2. Create a VM (e.g., Windows Server) and attach NICs:
 - WAN NIC → external switch (for public Internet).
 - LAN NIC → internal switch (for local IP range).
3. Install RRAS as a VPN and NAT server:
 - In Server Manager, add Remote Access role → enable Routing.
 - Launch RRAS console → configure Custom → select VPN (and NAT) roles.
 - Define WAN as the public interface and LAN as private interface for NAT routing, Microsoft Learn.
4. Configure Demand-Dial (Site-to-Site) VPN:
 - Create a new demand-dial interface with connection type Site-to-Site IPsec.

- Enter the public IPv4 address of the Azure VPN Gateway as remote endpoint.
- Use a pre-shared key—must match exactly what you'll configure in Azure.

5. Define IP addressing:

- LAN NIC with static IP.
- PPP/virtual interface will negotiate Azure Gateway.
- Ensure RRAS static routes to your LAN prefixes match between tunnel endpoints.



B. Azure Configuration

1. Create a Virtual Network & Gateway Subnet (/27 or larger).
2. Deploy an Azure VPN Gateway:
 - Choose route-based type.

- Recommended SKU.
- Optionally enable active-active mode if supported.

3. Create Local Network Gateway:

- Enter the on-prem RRAS public IP address.
- Define address prefixes for on-prem LAN networks.

4. Create the VPN Connection:

- Link Azure VPN Gateway and Local Network Gateway.
- Use the same pre-shared key used in RRAS.
- Verify connection type is Site-to-Site (IPsec/IKE).

5. Monitor deployment:

- Gateway creation may take 45 minutes or more.
- Connection status should show Connected once operational.

Create connection ...

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Virtual network gateway

To use a virtual network with a connection, it must be associated to a virtual network gateway. ⓘ

Virtual network gateway * ⓘ	VNet1GW
Local network gateway * ⓘ	Site1
Shared key (PSK) * ⓘ	 ✓
IKE Protocol ⓘ	☐ IKEv1 ☒ IKEv2
Use Azure Private IP Address ⓘ	☐
Enable BGP ⓘ	☐
FastPath ⓘ	☐
IPsec / IKE policy ⓘ	<input checked="" type="button" value="Default"/> Custom
Use policy based traffic selector ⓘ	Enable <input checked="" type="button" value="Disable"/>
DPD timeout in seconds * ⓘ	45 ✓
Connection Mode ⓘ	☒ Default ☐ InitiatorOnly ☐ ResponderOnly

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4. VERIFICATION & TESTING

1. Ping tests:

- From an Azure VM, ping the on-prem RRAS server's internal LAN IP.
- From an Azure VM ping a test machine behind RRAS.
- From an on-prem VM ping Azure VNet IP(s).
- Azure-to-on-prem ping may only reach RRAS control interface if routing not properly propagated.

2. Traceroute / Routing check:

- Use tracert to inspect path and confirm traffic flows through the tunnel.

3. RRAS interface status:

- Virtual adapter in RRAS console should be Connected.
- Check Event Viewer for IPsec/IKE negotiation logs.

5. SECURITY & ROUTING CONSIDERATIONS

- Ensure no overlapping IP ranges between on-prem LAN and Azure VNet—otherwise, Azure routing fails.
- Use strong pre-shared keys and preferably IKEv2 for better security.
- When using ExpressRoute and VPN, only advertise VPN endpoint IPs over ExpressRoute; route actual network prefixes via VPN to enforce encryption.
- Consider enabling Azure High Bandwidth tunnels (VpnGw5AZ, active-active, FastPath) for up to 20 Gbps throughput if needed.

CONCLUSION

This R&D demonstrated how to set up a secure Site-to-Site VPN between a Hyper-V-based on-prem environment and Azure using RRAS and VPN Gateway. The setup serves as a scalable foundation for enterprise or lab-based connectivity scenarios.
